Figure Captions and Notes

Figure 1: Pipeline Schematic

Suggested caption: "Overview of the 9-stage computational pipeline. Input is an Orphanet or OMIM identifier. Each stage queries public biological databases and applies scoring logic. Hard gates (orange diamonds) auto-reject or flag matches if critical thresholds are not met. Final output is a ranked list of surrogate-precedent gene therapy strategies with auto-generated protocols."

Figure 2: Multi-Dimensional Scoring Radar Chart

Suggested caption: "Multi-dimensional score profile for the top-matched prospective case study (CNS ultra-rare disease, Case Study 1). Each axis represents a normalized 0–1 score. The shaded area illustrates the composite area under the scoring profile. High platform depth and structural homology scores drive the top ranking despite moderate promoter match novelty."

Figure 3: Retrospective Validation Summary

Suggested caption: "Retrospective validation of the scoring engine against two approved gene therapies and one clinically discontinued program. Luxturna and Zolgensma achieve composite scores above the high-confidence threshold (dashed line), while the negative control (discontinued due to immunogenicity) is correctly down-ranked. Individual dimension contributions are shown as stacked segments."

Figure 4: Sensitivity Analysis Heatmap

Suggested caption: "Sensitivity of top-match ranking to $+/-20\%$ perturbations in individual scoring dimension weights. Color stability across a row indicates that a disease's top match is robust to weight variation. Structural homology and platform depth exert the strongest influence on rank stability for the tested retrospective and prospective cases."

Figure 5: Prospective Protocol Comparison

Suggested caption: "Auto-generated gene therapy protocols for the two prospective case studies. Left: CNS ultra-rare disease with intrathecal AAV9 delivery and ubiquitous promoter due to broad neuronal target requirement. Right: Metabolic hepatic disease with IV AAV8 delivery and liver-specific TBG promoter. Both protocols were generated in under two minutes and are consistent with clinical precedents for the respective tissues."

Supplementary Table 1: Full Surrogate Database

Supplementary Table 2: Complete Scoring Results

Supplementary Note 1: Weight Selection Justification